

ACTIVITY 0000

SCENARIO

During lunch break at Ala-Too many students buy several food items such as tea, samsa, salad, or rice meals. The cashier wants a simple program that can quickly calculate the total bill for one student and apply a discount when the order is large enough.

The program should:

- Read the number of items bought
- Read the price of each item
- Calculate the total bill
- Apply a 10% discount if the total is more than 500 soms
- Print the final amount to pay

Output:

Total before discount

Discount amount

Final payment

ACTIVITY 0001

SCENARIO

The Ala-Too library lends books to students for limited periods. Sometimes students return books late, and the librarian must calculate penalties depending on how many days each book is overdue. Doing this manually for many books takes time, so a program is needed.

The program should:

- Read the number of returned books
- Read the overdue days for each book
- Calculate the penalty for each book

RULES:

1. 15 soms for each overdue day
2. Extra 50 soms if a book is overdue by more than 10 days

Output:

Fine for each book
Total fine for all books

ACTIVITY 0002

SCENARIO

After a programming quiz, a teacher wants to quickly review class performance. Instead of checking results one by one, the teacher wants a program that can process all student scores and show the main statistics of the group.

The program should:

- Read the number of students
- Read each student's score
- Find the highest score
- Find the lowest score
- Calculate the average score
- Count how many students passed and failed

RULES:

Passing score is 50 or above

Output:

Highest score

Lowest score

Average score

Number of passed students

Number of failed students

ACTIVITY 0003

SCENARIO

At the university entrance, campus security checks cars entering the parking area. Some cars belong to students and teachers, while others belong to visitors. Security wants a basic program to separate vehicles that can enter directly from those that need manual approval.

The program should:

- Read the number of arriving cars
- Read the plate number of each car
- Read the driver type: student, teacher, or guest
- Separate cars into groups

RULES:

Students and teachers are allowed to enter

Guests must be marked for manual approval

Output:

list of allowed cars

list of guest cars

total number in each group

ACTIVITY 0004

SCENARIO

The dormitory administration checks electricity usage every month to see which rooms use normal amounts of energy and which rooms use too much.

A simple program is needed to process room data and prepare a basic report.

The program should:

- Reads the number of rooms
- Reads each room number
- Reads electricity usage for each room
- Marks the usage level of each room

RULES:

1. more than 300 kWh = High Usage
2. 300 kWh or less = Normal Usage

Output:

room number

electricity usage

usage status

total electricity usage

average usage

number of high usage rooms